EE1204: Engineering Electromagnetics — Marks: 80

Mid-Semester Exam: 14th Feb 2026 — Duration: 2 hours 30 minutes

Instructions:

- Marks will be awarded only if all assumptions (if any) are explicitly stated and clearly justified.
- If a question requires a supporting figure, marks will be awarded only if a neat, properly labeled figure is included.
- All 8 questions carry equal weightage: **10 marks each**.

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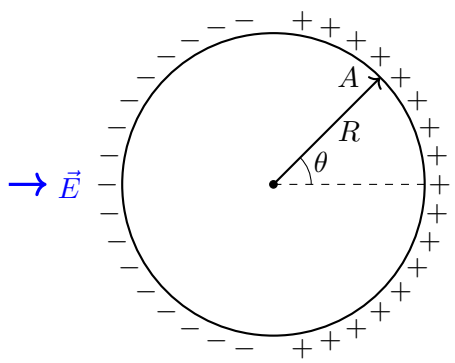
1. A solid conducting sphere of radius R is placed in a uniform electric field $\vec{E}$ as shown in Figure. Due to electric field non-uniform surface charges are induced on the surface of the sphere. Consider a point A on the surface of sphere at a polar angle θ from the direction of electric field as shown in Figure. Find the surface density of induced charges at point A in terms of electric field and polar angle θ.



Solution:

A neutral conducting sphere placed in a uniform external electric field $\vec{E}_0 = E_0\hat{z}$ develops induced surface charges. By symmetry, the induced charges are positive on one hemisphere and negative on the other, so the induced field has dipolar character.

The electric field outside the sphere can be written as the superposition of the applied field and the field of an induced dipole $\vec{p}$ at the center:

$$\vec{E} = \vec{E}_0 + \vec{E}_{\text{dip}}.$$

The electric field of a dipole is

$$\vec{E}_{\text{dip}} = \frac{1}{4\pi\varepsilon_0 r^3}\left(2p\cos\theta\,\hat{r} + p\sin\theta\,\hat{\theta}\right).$$

At the surface of a conductor, the tangential component of the electric field must vanish. The applied field has a tangential component

$$E_{0,\theta} = E_0\sin\theta,$$

so at $r = R$,

$$\frac{p \sin\theta}{4\pi\varepsilon_0 R^3} = E_0 \sin\theta.$$

This gives the induced dipole moment

$$p = 4\pi\varepsilon_0 R^3 E_0.$$

The radial component of the total electric field at the surface is then

$$E_r(R,\theta) = E_0 \cos\theta + \frac{2p\cos\theta}{4\pi\varepsilon_0 R^3} = 3E_0 \cos\theta.$$

For a conductor, the surface charge density is related to the normal field by

$$\sigma(\theta) = \varepsilon_0 E_r(R,\theta).$$

Hence,

$$\sigma(\theta) = 3\varepsilon_0 E_0 \cos\theta$$

2. A point charge of 20 nC is located at the origin. Four infinitely long uniform line charges are located parallel to the z-axis in the plane $x = 0$ as follows: Two line charges of density 80 nC/m at $y = -1m$ and -5 m, and two line charges of density -50 nC/m at $y = -2$ m and -4 m. Determine the total electric flux crossing the plane $y = +2$

Solution:

A point charge $q = 20$ nC is at the origin. Four infinitely long line charges (parallel to z-axis) lie at

$$(x,y) = (0,-1),(0,-5) \quad \text{with } \lambda = +80 \text{ nC/m}, \qquad (x,y) = (0,-2),(0,-4) \quad \text{with } \lambda = -50 \text{ nC/m}.$$

Let S be the plane $y = 2$ with unit normal $\hat{\mathbf{n}} = +\hat{\mathbf{y}}$. The electric flux of $\mathbf{E}$ through S is

$$\Phi_E = \iint_S \mathbf{E} \cdot d\mathbf{a} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} E_y(x,2,z)\, dx\, dz.$$

Contribution of the point charge

An infinite plane subtends a solid angle $\Omega = 2\pi$ at any point charge located on one side of it. Since the total flux from a point charge through a closed surface is q/ε_0, the fraction intercepted by the plane is $\Omega/(4\pi) = 1/2$. Hence

$$\Phi_{E,\text{point}} = \frac{q}{2\varepsilon_0}.$$

Numerically,

$$\Phi_{E,\text{point}} = \frac{20 \times 10^{-9}}{2\varepsilon_0} \approx 1.129 \times 10^3 \text{ N m}^2/\text{C}.$$

Derivation of $E_y(x, 2)$ for an infinite line charge at $(0, y_0)$

Consider an infinite line charge of density λ oriented along the z-axis and passing through the point $(x, y) = (0, y_0)$. Because the charge distribution is translationally invariant along z, the electric field is independent of z.

Field of an infinite line charge. In the plane transverse to the line (the $x-y$ plane), the field is purely radial (cylindrical symmetry):

$$\vec{E}(\rho) = \frac{\lambda}{2\pi\varepsilon_0}\,\frac{1}{\rho}\,\hat{\boldsymbol{\rho}},$$

where ρ is the perpendicular distance from the line to the observation point, and $\hat{\boldsymbol{\rho}}$ is the unit vector pointing away from the line in the $x-y$ plane.

Geometry on the plane $y = 2$. Take a field point on the plane $y = 2$: $(x, 2, z)$. The displacement from the line (in the transverse plane) is

$$\boldsymbol{\rho} = x\,\hat{\mathbf{x}} + (2 - y_0)\,\hat{\mathbf{y}}.$$

Define

$$a \equiv 2 - y_0,$$

so that $\boldsymbol{\rho} = x\,\hat{\mathbf{x}} + a\,\hat{\mathbf{y}}$. The distance is

$$\rho = \sqrt{x^2 + a^2},$$

and the corresponding unit vector is

$$\hat{\boldsymbol{\rho}} = \frac{\boldsymbol{\rho}}{\rho} = \frac{x\,\hat{\mathbf{x}} + a\,\hat{\mathbf{y}}}{\sqrt{x^2 + a^2}}.$$

Compute the y-component. Substitute into $\vec{E}(\rho)$:

$$\vec{E}(x, 2) = \frac{\lambda}{2\pi\varepsilon_0}\,\frac{1}{\sqrt{x^2 + a^2}}\left(\frac{x\,\hat{\mathbf{x}} + a\,\hat{\mathbf{y}}}{\sqrt{x^2 + a^2}}\right) = \frac{\lambda}{2\pi\varepsilon_0}\,\frac{x\,\hat{\mathbf{x}} + a\,\hat{\mathbf{y}}}{x^2 + a^2}.$$

Therefore the normal component to the plane $y = 2$ (i.e., the $\hat{\mathbf{y}}$ component) is

$$E_y(x, 2) = \frac{\lambda}{2\pi\varepsilon_0}\,\frac{a}{x^2 + a^2}, \qquad a = 2 - y_0.$$

Sign note. If $y_0 < 2$ then $a > 0$. For $\lambda > 0$, $E_y > 0$ (field points toward $+\hat{y}$); for $\lambda < 0$, $E_y < 0$.

Contribution of the infinite line charges

For a line charge λ located at $(0, y_0)$, the field is independent of z, and on the plane $y = 2$,

$$E_y(x, 2) = \frac{\lambda}{2\pi\varepsilon_0} \frac{a}{x^2 + a^2}, \qquad a = 2 - y_0 > 0.$$

Therefore the flux through a *strip* of the plane of length L_z (along z) is

$$\Phi_{E,\text{line}}(L_z) = \int_{-\infty}^{\infty} \int_0^{L_z} E_y(x, 2)\, dz\, dx = L_z \int_{-\infty}^{\infty} \frac{\lambda}{2\pi\varepsilon_0} \frac{a}{x^2 + a^2}\, dx.$$

Using $\displaystyle\int_{-\infty}^{\infty} \frac{a}{x^2 + a^2}\, dx = \pi$, we get

$$\Phi_{E,\text{line}}(L_z) = \frac{\lambda}{2\varepsilon_0} L_z.$$

Hence, for all four lines,

$$\lambda_{\text{tot}} = (80 + 80 - 50 - 50)\,\text{nC/m} = 60\ \text{nC/m},$$

$$\Phi_{E,\text{lines}}(L_z) = \frac{\lambda_{\text{tot}}}{2\varepsilon_0} L_z = \frac{60 \times 10^{-9}}{2\varepsilon_0} L_z \approx (3.388 \times 10^3)\, L_z\ \text{N m}^2/\text{C}.$$

Total

For a plane strip of length L_z along z,

$$\Phi_{E,\text{total}}(L_z) = \Phi_{E,\text{point}} + \Phi_{E,\text{lines}}(L_z) = \frac{1}{2\varepsilon_0}\Big(q + \lambda_{\text{tot}} L_z\Big)$$

Numerically,

$$\Phi_{E,\text{total}}(L_z) \approx 1.129 \times 10^3 + (3.388 \times 10^3)\, L_z \quad \text{N m}^2/\text{C}\ .$$

Important note: Because the line charges are infinitely long and the plane $y = 2$ is infinite in z, the *total* flux through the entire plane diverges (it grows proportional to L_z). The finite, meaningful result for the line charges is therefore the flux *per unit length* in z:

$$\frac{\Phi_{E,\text{lines}}}{L_z} = \frac{\lambda_{\text{tot}}}{2\varepsilon_0}$$

3. Consider the two scalar potential functions

$$V_1(x, y) = x \text{ and } V_2(x, y) = x + e^{-y} \sin x.$$

   (a) Show that both V_1 and V_2 satisfy Laplace's equation.

(b) Verify that both potentials satisfy the boundary condition $V = 0$ on the boundary $x = 0$.

(c) Are V_1 and V_2 identical everywhere in space? Justify your answer.

(d) Does the uniqueness theorem apply to this problem? Explain clearly which assumptions of the theorem are or are not satisfied.

Solution:

We are given the two potential functions

$$V_1(x, y) = x, \qquad V_2(x, y) = x + e^{-y} \sin x.$$

(a) **Verification of Laplace's equation**

Laplace's equation in two dimensions is

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2}.$$

For V_1,

$$\frac{\partial^2 V_1}{\partial x^2} = 0, \qquad \frac{\partial^2 V_1}{\partial y^2} = 0,$$

hence,

$$\nabla^2 V_1 = 0.$$

For V_2,

$$\frac{\partial^2}{\partial x^2}(e^{-y} \sin x) = -e^{-y} \sin x,$$

$$\frac{\partial^2}{\partial y^2}(e^{-y} \sin x) = e^{-y} \sin x.$$

Therefore,

$$\nabla^2 V_2 = -e^{-y} \sin x + e^{-y} \sin x = 0.$$

Thus, both V_1 and V_2 satisfy Laplace's equation.

(b) **Verification of boundary condition**

On the boundary $x = 0$,

$$V_1(0, y) = 0,$$

$$V_2(0, y) = 0 + e^{-y} \sin 0 = 0.$$

Hence both potentials satisfy the boundary condition

$$V = 0 \quad \text{on } x = 0.$$

(c) **Are V_1 and V_2 identical everywhere?**
The difference between the two potentials is

$$V_2 - V_1 = e^{-y} \sin x,$$

which is not identically zero.
Therefore,

$$V_1 \neq V_2$$

in general, even though both satisfy Laplace's equation and the same boundary condition on $x = 0$.

(d) **Applicability of the uniqueness theorem**
The uniqueness theorem for Laplace's equation requires that the potential be specified on a *closed boundary*, or that sufficient boundary conditions be provided to uniquely determine the solution.
In this problem, the potential is specified only on the line $x = 0$, which does not form a closed boundary. No conditions are imposed as $x \to \infty$ or $y \to \pm\infty$.
Therefore, the boundary conditions are incomplete and the uniqueness theorem does not apply. Hence, multiple solutions satisfying the given conditions are possible.

4. N charged raindrops, each of radius a, all have the same potential V. Assume the drops are far enough apart that the charge distribution on each drop is not affected by the others (i.e., each drop remains spherically symmetric).

   (a) Determine the total capacitance of the system.
   (b) If the N drops are combined into one big spherical drop (same material), find the capacitance of the big drop.
   (c) Compare the two capacitances.

Solution

a) Capacitance of one isolated spherical drop

Each raindrop is a conducting sphere of radius a, isolated in free space. For an isolated conducting sphere, the potential (with respect to infinity) is

$$V = \frac{1}{4\pi\varepsilon_0} \frac{q}{a}.$$

Hence,

$$q = 4\pi\varepsilon_0 a\, V.$$

By definition $C = q/V$, so the capacitance of one drop is

$$C_1 = 4\pi\varepsilon_0 a.$$

Total capacitance of N widely separated drops at the same potential

Because the drops are far apart, each drop behaves like an independent isolated sphere. If all drops are held at the same potential V, then each drop carries charge

$$q_i = C_1 V = (4\pi\varepsilon_0 a)V.$$

The total charge on the system is the sum:

$$Q_{\text{tot}} = \sum_{i=1}^{N} q_i = N(4\pi\varepsilon_0 a)V.$$

The equivalent (total) capacitance is defined by

$$C_{\text{tot}} = \frac{Q_{\text{tot}}}{V}.$$

Therefore,

$$C_{\text{tot}} = N\, 4\pi\varepsilon_0 a.$$

b) Capacitance after combining the drops into one big drop

When N identical spherical drops coalesce into one larger spherical drop, the total volume is conserved:

$$N\left(\frac{4}{3}\pi a^3\right) = \frac{4}{3}\pi R^3,$$

so the radius R of the big drop is

$$R = a\, N^{1/3}.$$

The capacitance of an isolated conducting sphere of radius R is

$$C_{\text{big}} = 4\pi\varepsilon_0 R = 4\pi\varepsilon_0 a\, N^{1/3}.$$

Hence,

$$C_{\text{big}} = 4\pi\varepsilon_0 a\, N^{1/3}.$$

c) Comparison

Compare the two results:

$$C_{\text{tot}} = 4\pi\varepsilon_0 a\, N, \qquad C_{\text{big}} = 4\pi\varepsilon_0 a\, N^{1/3}.$$

Therefore,

$$\frac{C_{\text{tot}}}{C_{\text{big}}} = \frac{N}{N^{1/3}} = N^{2/3}.$$

So for $N > 1$,

$$C_{\text{tot}} > C_{\text{big}}, \text{ and the separated system has } N^{2/3} \text{ times larger capacitance.}$$

Physical remark (why this makes sense): Capacitance of an isolated conductor scales with its linear size ($C \propto R$). If you merge droplets, the radius grows only as $N^{1/3}$, so the capacitance increases slowly. But with N separated droplets all at the same potential, the total charge stored is the sum over all droplets, so the equivalent capacitance adds roughly linearly in N.

5. A hollow conducting spherical shell of radius R is given a total charge Q, which finally resides uniformly on its surface. Imagine building up the final charge by successively adding infinitesimally thin shells of charge dq. When the charge already on the shell is q, find the work dW required to bring an additional charge dq from infinity and place it on the shell. Hence determine the total energy U stored in the charged shell.

Solution

Key idea: work to add charge = (existing potential) × (added charge)

At any intermediate stage of the charging process, suppose the spherical conductor already carries charge q. Since it is a conductor in electrostatic equilibrium, all charge resides on the outer surface and the potential of the entire conductor is uniform.

If we now bring a small additional charge dq from infinity (where we take potential to be zero) to the shell, the required incremental work is

$$dW = V(q)\, dq,$$

where $V(q)$ is the potential of the shell when its charge is q.

So the problem reduces to finding $V(q)$.

Potential of a conducting sphere of radius R carrying charge q

For a conducting sphere (or spherical shell) of radius R with charge q, the electric field outside is the same as that of a point charge q at the center:

$$E(r) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}, \qquad r \geq R,$$

and inside the conductor (and inside the hollow cavity) the field is zero.

The potential of the shell relative to infinity is

$$V(q) = \phi(R) - \phi(\infty) = \int_{\infty}^{R} \mathbf{E} \cdot d\mathbf{l} = \int_{\infty}^{R} \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\, dr.$$

Compute the integral:

$$V(q) = \frac{q}{4\pi\varepsilon_0}\left[-\frac{1}{r}\right]_{\infty}^{R} = \frac{q}{4\pi\varepsilon_0 R}.$$

Incremental work to add dq

Substitute $V(q)$ into $dW = V(q)\, dq$:

$$dW = \frac{q}{4\pi\varepsilon_0 R}\, dq.$$

Total energy stored

To build the charge from 0 up to Q, integrate dW from $q = 0$ to $q = Q$:

$$U = \int_0^Q dW = \int_0^Q \frac{q}{4\pi\varepsilon_0 R}\, dq = \frac{1}{4\pi\varepsilon_0 R}\int_0^Q q\, dq.$$

Evaluate:

$$U = \frac{1}{4\pi\varepsilon_0 R}\left[\frac{q^2}{2}\right]_0^Q = \frac{Q^2}{8\pi\varepsilon_0 R}.$$

Optional check (connect to capacitance)- No marks for this: The capacitance of an isolated conducting sphere is $C = 4\pi\varepsilon_0 R$. Hence $U = \dfrac{Q^2}{2C} = \dfrac{1}{2}QV$, consistent with the standard energy formula for a capacitor.

6. (a) At each corner of a square of side a there is a particle with charge q. A point charge of opposite sign and magnitude Q is fixed at the center of the square. What value of Q makes the *net electrostatic force* on *each* corner charge equal to zero?

   (b) With Q chosen as in part (a), show that the *total electrostatic potential energy* of the system is zero.

Solution

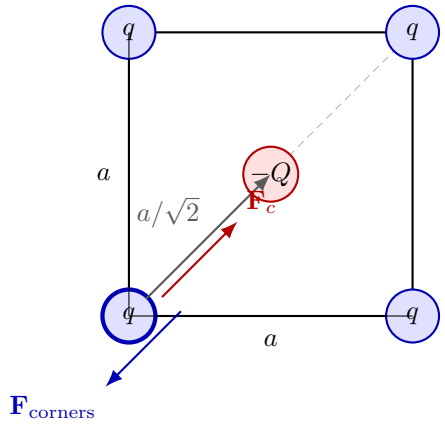


Figure 1: Square of corner charges q and a central charge $-Q$ (opposite sign). The chosen corner (thicker outline) experiences $\mathbf{F}_{\text{corners}}$ along $-(\hat{x}+\hat{y})$ and $\mathbf{F}_c$ toward the center.

(a) Choose Q so the net force on a corner charge is zero

Pick the corner charge at the origin $(0,0)$. Let the square corners be at

$$(0,0),\ (a,0),\ (0,a),\ (a,a),$$

and the center is at $\left(\frac{a}{2},\frac{a}{2}\right)$. All corner charges are q. The central charge is $-Q$ if $q>0$ ("opposite sign").

Forces on the corner at $(0,0)$ due to the other three corner charges:

- From the charge at $(a,0)$: separation a, repulsive along $-\hat{\mathbf{x}}$,

$$\mathbf{F}_{(a,0)\to(0,0)} = -\,\hat{\mathbf{x}}\,\frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}.$$

- From the charge at $(0,a)$: separation a, repulsive along $-\hat{\mathbf{y}}$,

$$\mathbf{F}_{(0,a)\to(0,0)} = -\,\hat{\mathbf{y}}\,\frac{1}{4\pi\varepsilon_0}\frac{q^2}{a^2}.$$

- From the charge at (a,a): separation $a\sqrt{2}$, repulsive along the diagonal $-(\hat{\mathbf{x}}+\hat{\mathbf{y}})/\sqrt{2}$,

$$\mathbf{F}_{(a,a)\to(0,0)} = -\frac{\hat{\mathbf{x}}+\hat{\mathbf{y}}}{\sqrt{2}}\,\frac{1}{4\pi\varepsilon_0}\frac{q^2}{(a\sqrt{2})^2} = -\frac{\hat{\mathbf{x}}+\hat{\mathbf{y}}}{\sqrt{2}}\,\frac{1}{4\pi\varepsilon_0}\frac{q^2}{2a^2}.$$

Add components. Let $k = \frac{1}{4\pi\varepsilon_0}$. Then

$$F_x^{(\text{corners})} = -k\frac{q^2}{a^2} \; - \; k\frac{q^2}{2a^2}\frac{1}{\sqrt{2}}, \qquad F_y^{(\text{corners})} = -k\frac{q^2}{a^2} \; - \; k\frac{q^2}{2a^2}\frac{1}{\sqrt{2}}.$$

So the resultant from the three corner charges points along $-(\hat{\mathbf{x}} + \hat{\mathbf{y}})$ and has equal x and y components.

Force due to the central charge:

Distance from center to a corner is

$$r_c = \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2} = \frac{a}{\sqrt{2}}.$$

Magnitude of attraction between q and $-Q$ is

$$F_c = k\frac{|qQ|}{r_c^2} = k\frac{qQ}{(a/\sqrt{2})^2} = k\frac{2qQ}{a^2}.$$

Direction is from the corner toward the center: $+(\hat{\mathbf{x}} + \hat{\mathbf{y}})/\sqrt{2}$. Hence

$$\mathbf{F}_c = \frac{\hat{\mathbf{x}} + \hat{\mathbf{y}}}{\sqrt{2}}\, k\frac{2qQ}{a^2}.$$

So each component is

$$F_{cx} = F_{cy} = k\frac{2qQ}{a^2}\frac{1}{\sqrt{2}} = k\frac{\sqrt{2}\,qQ}{a^2}.$$

Net force zero condition: require $F_x = 0$ (and automatically $F_y = 0$):

$$k\frac{\sqrt{2}\,qQ}{a^2} = k\frac{q^2}{a^2} + k\frac{q^2}{2a^2}\frac{1}{\sqrt{2}}.$$

Cancel k/a^2:

$$\sqrt{2}\,qQ = q^2\left(1 + \frac{1}{2\sqrt{2}}\right).$$

Solve for Q:

$$Q = \frac{q}{\sqrt{2}}\left(1 + \frac{1}{2\sqrt{2}}\right) = q\left(\frac{1}{\sqrt{2}} + \frac{1}{4}\right).$$

$$Q = q\left(\frac{1}{\sqrt{2}} + \frac{1}{4}\right), \qquad \text{and its sign is opposite to } q.$$

So the central charge is $-Q$ if the corner charges are $+q$.

(b) Total potential energy is zero for this value of Q

The total electrostatic potential energy is the sum over all distinct pairs:

$$U = \sum_{\text{pairs } (i<j)} k\frac{q_i q_j}{r_{ij}}.$$

There are two types of interactions:

1) Corner–corner pairs: 4 corner charges give $\binom{4}{2} = 6$ pairs. Among these:

- 4 edges of length a: energy per pair $k\frac{q^2}{a}$, total $U_{\text{edge}} = 4k\frac{q^2}{a}$.
- 2 diagonals of length $a\sqrt{2}$: energy per pair $k\frac{q^2}{a\sqrt{2}}$, total $U_{\text{diag}} = 2k\frac{q^2}{a\sqrt{2}} = k\frac{\sqrt{2}\,q^2}{a}$.

So

$$U_{\text{corners}} = k\frac{q^2}{a}\left(4+\sqrt{2}\right).$$

2) Center–corner pairs: 4 pairs, each at distance $a/\sqrt{2}$, with charges q and $-Q$. Each contributes

$$k\frac{q(-Q)}{a/\sqrt{2}} = -k\frac{\sqrt{2}\,qQ}{a}.$$

So

$$U_{\text{center}} = 4\left(-k\frac{\sqrt{2}\,qQ}{a}\right) = -k\frac{4\sqrt{2}\,qQ}{a}.$$

Hence total energy

$$U = k\frac{1}{a}\left[q^2(4+\sqrt{2}) - 4\sqrt{2}\,qQ\right].$$

Now substitute $Q = q\left(\frac{1}{\sqrt{2}} + \frac{1}{4}\right)$:

$$4\sqrt{2}\,qQ = 4\sqrt{2}\,q^2\left(\frac{1}{\sqrt{2}} + \frac{1}{4}\right) = 4q^2 + \sqrt{2}\,q^2.$$

Therefore,

$$U = k\frac{1}{a}\left[q^2(4+\sqrt{2}) - q^2(4+\sqrt{2})\right] = 0.$$

$$U = 0.$$

7. Two electric dipoles are located at the same point (the origin). Each has dipole moment magnitude p. One dipole moment points along $+\hat{\mathbf{x}}$, and the other points along $+\hat{\mathbf{y}}$. Determine the net (equivalent) dipole moment of the combined system (magnitude and direction).

Solution:

Key idea

For any localized charge distribution with net charge zero, the dipole moment is

$$\mathbf{p} = \sum_i q_i \, \mathbf{r}_i,$$

and if the system is a collection of dipoles (with no net charge), the total dipole moment is simply the *vector sum*:

$$\mathbf{p}_{\text{net}} = \mathbf{p}_1 + \mathbf{p}_2.$$

Here,

$$\mathbf{p}_1 = p\,\hat{\mathbf{x}}, \qquad \mathbf{p}_2 = p\,\hat{\mathbf{y}}.$$

Therefore,

$$\mathbf{p}_{\text{net}} = p\,\hat{\mathbf{x}} + p\,\hat{\mathbf{y}}.$$

Magnitude and direction

The magnitude is

$$|\mathbf{p}_{\text{net}}| = \sqrt{p^2 + p^2} = \sqrt{2}\,p.$$

The direction is along $\hat{\mathbf{x}} + \hat{\mathbf{y}}$, i.e., at 45° above the $+x$-axis in the xy-plane:

$$\hat{\mathbf{p}}_{\text{net}} = \frac{\hat{\mathbf{x}} + \hat{\mathbf{y}}}{\sqrt{2}} \ .$$

(Optional) Check using explicit charges

Represent each dipole using charges $\pm q$ separated by distance d, so $p = qd$. Place charges as follows (all in the plane $z = 0$):

$$+q \text{ at } \left(\frac{d}{2}, 0\right), \quad -q \text{ at } \left(-\frac{d}{2}, 0\right) \quad \Rightarrow \quad \mathbf{p}_1 = qd\,\hat{\mathbf{x}} = p\,\hat{\mathbf{x}},$$

$$+q \text{ at } \left(0, \frac{d}{2}\right), \quad -q \text{ at } \left(0, -\frac{d}{2}\right) \quad \Rightarrow \quad \mathbf{p}_2 = qd\,\hat{\mathbf{y}} = p\,\hat{\mathbf{y}}.$$

Adding gives the same result $\mathbf{p}_{\text{net}} = p(\hat{\mathbf{x}} + \hat{\mathbf{y}})$.

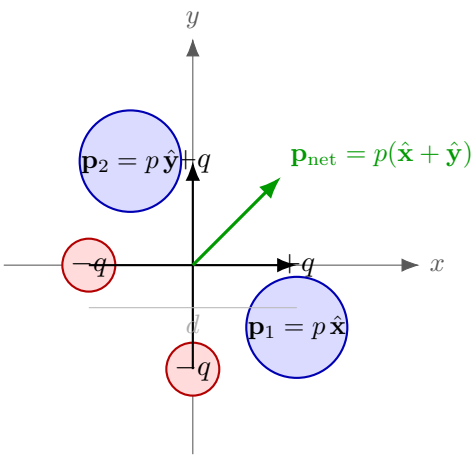


Figure 2: Two perpendicular dipoles in the $z = 0$ plane, each represented by charges $\pm q$ separated by d, so $p = qd$. The dipole moment points from $-q$ to $+q$; the net moment is the vector sum.

8. Consider a capacitor made of four large, parallel conducting plates (area A), separated by equal small gaps s (so the spacing between plate 1–2, 2–3, and 3–4 is s). Plates 1 and 3 are connected together by a wire, and plates 2 and 4 are connected together by a wire. Assuming fringing is negligible ($A \gg s^2$), determine the equivalent capacitance between the two wired terminals $(1, 3)$ and $(2, 4)$.

   Solution.

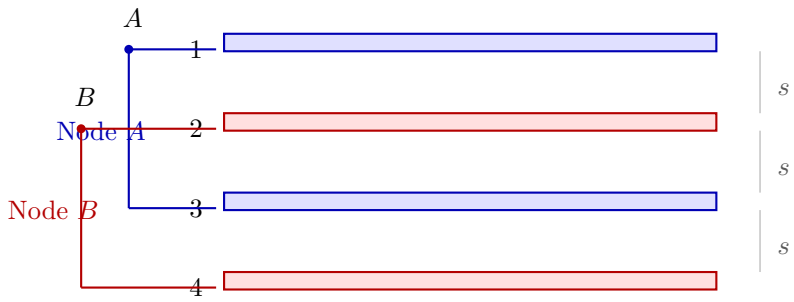


Figure 3: Four parallel plates with equal gaps s. Plates $1, 3$ are shorted (node A, blue) and plates $2, 4$ are shorted (node B, red). The three gaps each form a capacitor between the same two nodes $\Rightarrow$ parallel combination.

Let the two wired terminals be:

$$\text{Node } A: \ (1,3) \ \text{ at potential } V_A, \qquad \text{Node } B: \ (2,4) \ \text{ at potential } V_B, \qquad V \equiv V_A - V_B.$$

Because plates 1 and 3 are shorted, they are equipotential at V_A. Similarly plates 2 and 4 are equipotential at V_B.

Step 1: Capacitance of one gap (parallel-plate from Gauss law).

Consider one gap of thickness s between two large plates carrying surface charge densities $+\sigma$ and $-\sigma$. Using a pillbox Gaussian surface straddling one conducting surface,

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\varepsilon_0} \quad \Rightarrow \quad E = \frac{\sigma}{\varepsilon_0}$$

in the region between plates (uniform field; outside field cancels for large plates).

The potential difference across the gap is

$$V = \int_0^s E\, dl = Es = \frac{\sigma}{\varepsilon_0} s.$$

Charge on one plate face is $Q = \sigma A$. Hence the capacitance of a single gap is

$$C_0 = \frac{Q}{V} = \frac{\sigma A}{(\sigma s/\varepsilon_0)} = \frac{\varepsilon_0 A}{s}.$$

Step 2: Apply to the four-plate connection.

There are three gaps: $(1-2)$, $(2-3)$, and $(3-4)$, each of thickness s. In every gap, one plate belongs to node A and the other belongs to node B:

$$(1-2): \ A\text{–}B, \qquad (2-3): \ B\text{–}A, \qquad (3-4): \ A\text{–}B.$$

Since the two nodes are fixed at potentials V_A and V_B, the magnitude of the voltage across *each* gap is the same:

$$\Delta V_{12} = V_A - V_B = V, \quad \Delta V_{23} = V_B - V_A = -V \text{ (same magnitude)}, \quad \Delta V_{34} = V_A - V_B = V.$$

Therefore, each gap behaves as a capacitor $C_0 = \varepsilon_0 A/s$ connected between the same two terminals A and B. So the three capacitors are *in parallel*.

Total charge drawn from node A:

$$Q_A = Q_{12} + Q_{23} + Q_{34} = C_0 V + C_0 V + C_0 V = 3C_0 V.$$

Hence the equivalent capacitance is

$$C_{\text{eq}} = \frac{Q_A}{V} = 3C_0 = \frac{3\varepsilon_0 A}{s}.$$

Answer:

$$C_{\text{eq}} = \frac{3\varepsilon_0 A}{s}$$

(If the gaps are filled with a dielectric of permittivity ε, replace ε_0 by ε.)